MSG No. 95.140 P_c4_322 [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2'2'2.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,12,14]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,8,11,15]`
3	$\left[0, 0, \frac{1}{4}\right]$	`[3,7,10,16]`
4	$\left[0, 0, \frac{1}{2}\right]$	`[4,6,9,13]`

Table 2: Wyckoff site: `4b`, site symmetry: `2'2'2.`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,5,12,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,8,11,15]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[3,7,10,16]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,6,9,13]`

Table 3: Wyckoff site: `4c`, site symmetry: `2'2'2.`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,5,12,14]`
2	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[2,8,11,15]`
3	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[3,7,10,16]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,6,9,13]`

Table 4: Wyckoff site: `4d`, site symmetry: `2'22'.`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[1,4,13,14]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,7,11,16]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,8,10,15]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[5,6,9,12]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.22`

No.	position	mapping
1	$\left[0, 0, \frac{1}{8}\right]$	`[1,7,14,16]`
2	$\left[0, 0, \frac{7}{8}\right]$	`[2,5,11,12]`
3	$\left[0, 0, \frac{3}{8}\right]$	`[3,4,10,13]`
4	$\left[0, 0, \frac{5}{8}\right]$	`[6,8,9,15]`

Table 6: Wyckoff site: `4f`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{8}\right]$	`[1,7,14,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{7}{8}\right]$	`[2,5,11,12]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{8}\right]$	`[3,4,10,13]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{5}{8}\right]$	`[6,8,9,15]`

Table 7: Wyckoff site: `8g`, site symmetry: `2'..`

No.	position	mapping
1	$[0, 0, z]$	`[1,14]`
2	$\left[0, 0, z + \frac{3}{4}\right]$	`[2,11]`
3	$\left[0, 0, z + \frac{1}{4}\right]$	`[3,10]`
4	$\left[0, 0, \frac{1}{2} - z\right]$	`[4,13]`
5	$[0, 0, -z]$	`[5,12]`
6	$\left[0, 0, z + \frac{1}{2}\right]$	`[6,9]`
7	$\left[0, 0, \frac{1}{4} - z\right]$	`[7,16]`
8	$\left[0, 0, \frac{3}{4} - z\right]$	`[8,15]`

Table 8: Wyckoff site: `8h`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{3}{4}\right]$	`[2,11]`
3	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{4}\right]$	`[3,10]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[4,13]`
5	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5,12]`
6	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[6,9]`
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4} - z\right]$	`[7,16]`
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4} - z\right]$	`[8,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,14]`
2	$\left[0, \frac{1}{2}, z + \frac{3}{4}\right]$	`[2,11]`
3	$\left[0, \frac{1}{2}, z + \frac{1}{4}\right]$	`[3,10]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	`[4,13]`
5	$\left[\frac{1}{2}, 0, -z\right]$	`[5,12]`
6	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[6,9]`
7	$\left[0, \frac{1}{2}, \frac{1}{4} - z\right]$	`[7,16]`
8	$\left[0, \frac{1}{2}, \frac{3}{4} - z\right]$	`[8,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `.2.`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,5]`
2	$[-y,\ 0,\ \frac{3}{4}]$	`[2,8]`
3	$[y,\ 0,\ \frac{1}{4}]$	`[3,7]`
4	$[0,\ -y,\ \frac{1}{2}]$	`[4,6]`
5	$[0,\ y,\ \frac{1}{2}]$	`[9,13]`
6	$[-y,\ 0,\ \frac{1}{4}]$	`[10,16]`
7	$[y,\ 0,\ \frac{3}{4}]$	`[11,15]`
8	$[0,\ -y,\ 0]$	`[12,14]`

Table 11: Wyckoff site: `8k`, site symmetry: `.2'.`

No.	position	mapping
1	$[\frac{1}{2},\ y,\ \frac{1}{4}]$	`[1,13]`
2	$[-y,\ \frac{1}{2},\ 0]$	`[2,16]`
3	$[y,\ \frac{1}{2},\ \frac{1}{2}]$	`[3,15]`
4	$[\frac{1}{2},\ -y,\ \frac{1}{4}]$	`[4,14]`
5	$[\frac{1}{2},\ y,\ \frac{3}{4}]$	`[5,9]`
6	$[\frac{1}{2},\ -y,\ \frac{3}{4}]$	`[6,12]`
7	$[y,\ \frac{1}{2},\ 0]$	`[7,11]`
8	$[-y,\ \frac{1}{2},\ \frac{1}{2}]$	`[8,10]`

Table 12: Wyckoff site: `8l`, site symmetry: `.2'.`

No.	position	mapping
1	$[0,\ y,\ \frac{1}{4}]$	`[1,13]`
2	$[-y,\ 0,\ 0]$	`[2,16]`
3	$[y,\ 0,\ \frac{1}{2}]$	`[3,15]`
4	$[0,\ -y,\ \frac{1}{4}]$	`[4,14]`
5	$[0,\ y,\ \frac{3}{4}]$	`[5,9]`
6	$[0,\ -y,\ \frac{3}{4}]$	`[6,12]`
7	$[y,\ 0,\ 0]$	`[7,11]`
8	$[-y,\ 0,\ \frac{1}{2}]$	`[8,10]`

Table 13: Wyckoff site: `8m`, site symmetry: `.2.`

No.	position	mapping
1	$[\frac{1}{2},\ y,\ 0]$	`[1,5]`
2	$[-y,\ \frac{1}{2},\ \frac{3}{4}]$	`[2,8]`
3	$[y,\ \frac{1}{2},\ \frac{1}{4}]$	`[3,7]`
4	$[\frac{1}{2},\ -y,\ \frac{1}{2}]$	`[4,6]`
5	$[\frac{1}{2},\ y,\ \frac{1}{2}]$	`[9,13]`
6	$[-y,\ \frac{1}{2},\ \frac{1}{4}]$	`[10,16]`
7	$[y,\ \frac{1}{2},\ \frac{3}{4}]$	`[11,15]`

continued ...

Table 13

No.	position	mapping
8	$\left[\frac{1}{2}, -y, 0\right]$	[12,14]

Table 14: Wyckoff site: 8n, site symmetry: ..2'

No.	position	mapping
1	$\left[x, -x, \frac{1}{8}\right]$	[1,16]
2	$\left[x, x, \frac{7}{8}\right]$	[2,12]
3	$\left[-x, -x, \frac{3}{8}\right]$	[3,13]
4	$\left[x, x, \frac{3}{8}\right]$	[4,10]
5	$\left[-x, -x, \frac{7}{8}\right]$	[5,11]
6	$\left[-x, x, \frac{5}{8}\right]$	[6,15]
7	$\left[-x, x, \frac{1}{8}\right]$	[7,14]
8	$\left[x, -x, \frac{5}{8}\right]$	[8,9]

Table 15: Wyckoff site: 8o, site symmetry: ..2

No.	position	mapping
1	$\left[x, -x, \frac{3}{8}\right]$	[1,8]
2	$\left[x, x, \frac{1}{8}\right]$	[2,4]
3	$\left[-x, -x, \frac{5}{8}\right]$	[3,5]
4	$\left[-x, x, \frac{7}{8}\right]$	[6,7]
5	$\left[x, -x, \frac{7}{8}\right]$	[9,16]
6	$\left[x, x, \frac{5}{8}\right]$	[10,12]
7	$\left[-x, -x, \frac{1}{8}\right]$	[11,13]
8	$\left[-x, x, \frac{3}{8}\right]$	[14,15]

Table 16: Wyckoff site: 16p, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[-y, x, z + \frac{3}{4}\right]$	[2]
3	$\left[y, -x, z + \frac{1}{4}\right]$	[3]
4	$\left[x, -y, \frac{1}{2} - z\right]$	[4]
5	$[-x, y, -z]$	[5]
6	$\left[-x, -y, z + \frac{1}{2}\right]$	[6]
7	$\left[y, x, \frac{1}{4} - z\right]$	[7]
8	$\left[-y, -x, \frac{3}{4} - z\right]$	[8]
9	$\left[x, y, z + \frac{1}{2}\right]$	[9]
10	$\left[-y, x, z + \frac{1}{4}\right]$	[10]
11	$\left[y, -x, z + \frac{3}{4}\right]$	[11]
12	$[x, -y, -z]$	[12]
13	$\left[-x, y, \frac{1}{2} - z\right]$	[13]
14	$[-x, -y, z]$	[14]

continued ...

Table 16

No.	position	mapping
15	$\left[y,\, x,\, \frac{3}{4} - z\right]$	[15]
16	$\left[-y,\, -x,\, \frac{1}{4} - z\right]$	[16]